Yiquan Wang

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Research Interests: Quantitative Biology, Complex Systems, Network Robustness, Statistical Physics

Memberships: Biophysical Society of China • Chinese Chemical Society • IEEE Biometrics Council

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Education

Xinjiang University (National Base for Mathematical Research and Teaching Talents) Urumqi, Xinjiang
Bachelor of Science in Mathematics and Applied Mathematics 2023.09 – 2027.06

Tsinghua University & X-Institute Shenzhen, Guangdong
Tsien Excellence in Engineering Program (Joint Program, X Scholar) 2024.06 – 2027.06

Research Experience

First-Principles of Protein Geometry via Mathematical Physics (Independent Research) 2025.01 – Present
Researcher | Mathematical Physics & Discrete Differential Geometry

- Abandoned empirical force fields and AI-driven statistical fitting to model protein backbone geometry exclusively through the **discrete Hasimoto map** and **nonlinear Schrödinger equations**.
- Derived exact analytical solutions for local helical transitions, demonstrating sub-Angstrom precision. Proved high cooperativity in phase transitions from a pure kinematic perspective, aligning with the Zimm-Bragg model.

AI-Driven Biomolecular Dynamics & Representation Learning (Tsinghua University ESRT) 2024.08 – Present
Project Lead | Supervised by Prof. Kai Wei

- **Protein Dynamics Analysis:** Developed a sequence-context-aware decoding framework to robustly reconstruct protein conformational dynamics (thermal fluctuations) directly from crystallographic B-factors.
- **Topological Sequence Representation:** Proposed biological sequence "sonification" (1D to 2D spectrogram mapping), uncovering hidden frequency/topological features to predict protein function, achieving 90.44% accuracy on the CARE enzyme benchmark (**Published in JCIM**).

Network Topology & System Robustness (Provincial Innovation Program) 2024.03 – 2025.06
Project Lead | Supervised by Prof. Eminjan Sabir

- Investigated the **robustness and fault tolerance** of large-scale topology networks (k-ary n-cubes) under the Region-Based Fault (RBF) model, moving beyond unrealistic assumptions of independent random noise.
- Mathematically proved Hamiltonian connectivity under clustered failure conditions, demonstrating a deep understanding of network resilience and topological invariants in spatially correlated noisy environments.

Macro-System Dynamics & Cascading Failures (CAS Innovation Program) 2024.11 – 2025.09
Project Lead | Supervised by Researcher Yong Ge

- Constructed a high-fidelity Knowledge Graph (70,297 nodes) to decouple the **emergence and cascading failures** of compound heatwaves across interconnected physical, biological, and socio-economic networks.
- Identified bio-ecological systems as "non-linear amplifiers" of thermal stress, showcasing the ability to abstract and analyze the dynamics of multi-layered complex systems.

Shenzhen Bay Laboratory Shenzhen, Guangdong
Visiting Student | Institute of Neurological and Psychiatric Disorders (Wen Yuan's Group) 2025.07 – 2025.09

- Investigating the systemic dysregulation of gene regulatory networks in neurological disorders using high-throughput single-cell multi-omics and CRISPR screening.

Selected Publications

Group 1: Mathematical Physics & First-Principles of Biology

[1] **Wang, Y.*** (2026). Spectral entropy of the discrete Hasimoto effective potential exposes sub-residue geometric transitions in protein secondary structure. *arXiv preprint arXiv:2602.21787*. (Sole & corresponding author).
[2] **Wang, Y.*** (2026). Piecewise integrability of the discrete Hasimoto map for analytic prediction and design of helical peptides. *arXiv preprint arXiv:2602.16255*. (Sole & corresponding author).
[3] **Wang, Y.*** (2026). Structural barriers of the discrete Hasimoto map applied to protein backbone geometry. *arXiv preprint arXiv:2602.13160*. (Sole & corresponding author).

Group 2: Biomolecular Dynamics & Representation Learning (Showcasing Data & Engineering Capability)

[4] **Wang, Y.**, Cai, M., Dong, Y., Ma, Y., & Wei, K. (2025). From Signal to Symphony: Exploring 2D Sequence Representations for Protein Function Prediction. *Journal of Chemical Information and Modeling*, 65(23), 12723-12736. (**Top**).

[5] **Wang, Y.***, Cai, M., Ma, Y., & Wei, K. (2026). Sequence-context-aware decoding enables robust reconstruction of protein dynamics from crystallographic B-factors. *Learning Meaningful Representations of Life (LMRL) Workshop at ICLR 2026*.

[6] **Wang, Y.**, Ma, Y., Chang, Y., Yan, J., Zhang, J., Cai, M., & Wei, K. (2025). Diffusion Models at the Drug Discovery Frontier: A Review on Generating Small Molecules Versus Therapeutic Peptides. *Biology*, 14(12), 1665.

[7] Wang, X.†, **Wang, Y.**†, Huang, T.-Y., et al. (2025). Octopus Inspired Optimization: A Hierarchical Framework for Navigating Protein Fitness Landscapes. *2025 IEEE BIBM*, 6897-6904. (Co-first).

Group 3: Complex Systems, Network Topology & Robustness

[8] **Wang, Y.**, Liu, Z., Zai, J., Chen, J., & Sabir, E. (2026). Resilient AI Infrastructure by Design: A Spatially-Aware Framework for Tolerating Clustered Failures. *4th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE)*.

[9] **Wang, Y.***, Huang, T. Y., Gao, Q., Chang, Y., & Zhang, J. (2026). AI Driven Discovery of Bio Ecological Mediation in Cascading Heatwave Risks. *ICLR 2026 Workshop on Foundation Models for Science (FM4Science)*.

[10] **Wang, Y.***, Chang, Y., Zhang, J., et al. (2026). Geometric Fragility in Alzheimer's Disease: Probing the Loss of Hippocampal Hierarchical Abstraction via Contrastive Point Cloud Modeling. *LMRL Workshop at ICLR 2026*.

Professional Skills & Academic Engagement

- **Technical Skills:** Network Analysis (Graph Theory), Dynamical Systems Modeling, Stochastic Processes, Python, C/C++, MATLAB, Linux, Multi-omics, Molecular Dynamics Simulation.
- **Academic Engagement:** Tsinghua-Peking University Center for Life Sciences Summer Camp (2025); Brown University Department of Physics Artificial Intelligence Winter School (2025); Fudan University Summer School of Mathematical Logic (2024); Wuhan University National Tianyuan Mathematics Center Discussion Class (2024).
- **Selected Awards:** Synthetic Biology Challenges Silver Award $\times$ 2 (2025); Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM) National Second Prize (2025).